CLOUD SECURITY SPECIALIST

NAME:KEMUNTO FAITH OGWOKA
CS NO:ADC-CSS02-25071
GMAIL:faithogwoka@gmail.com

LAB:SECURITY FOR VIRTUAL NETWORKS

1.1 INTRODUCTION

I started the lab with a required lab set up.I clicked on ctrl alt and delete to bring up the login page,signed in as admin and then downloaded the lab files.

1.2 LAB 02:NETWORK SECURITY GROUPS AND APPLICATION SECURITY GROUPS

LAB SCENARIO

In this lab,I created the virtual networking infrastructure, deployed virtual machines and tested the network filters.

EXERCISE 1:CREATE THE VIRTUAL NETWORKING INFRASTRUCTURE

In this exercise, I created a virtual network with one subnet,two application security groups,a network security group and associated it with the virtual network subnet and inbound NSG security rules to all traffic to web servers and RDP to the management servers.

TASK 1

In this task, I created a virtual network to use with the network and application security groups.

Step 1

I signed in to the Azure Portal with the credentials provided for the lab

Step 2

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Virtual networks and pressed the Enter key

Step 3

On the Virtual networks blade, I clicked on + Create

Step 4

On the Basics tab of the Create virtual network blade, I specified the settings provided, left others with their default values and clicked Next: IP Addresses:

Step 5

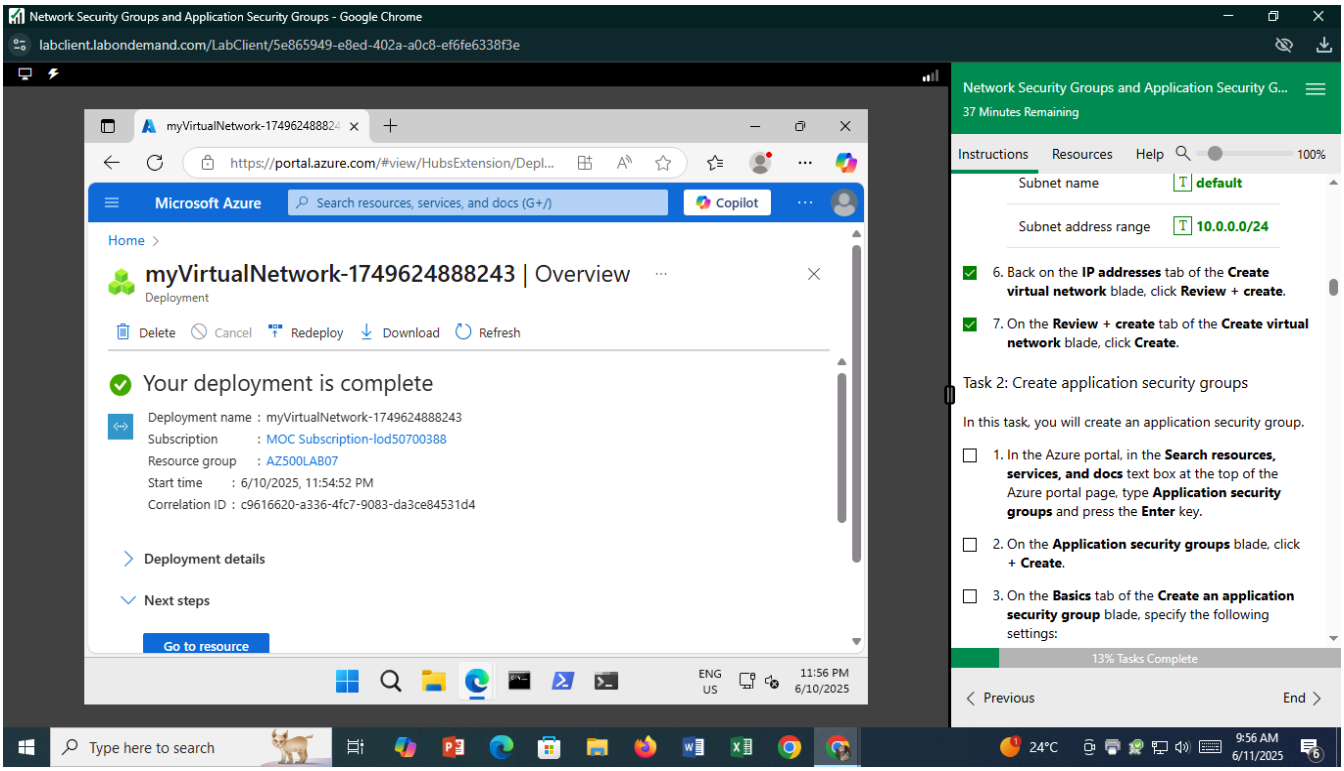
On the IP addresses tab of the Create virtual network blade, I set the IPv4 address space to 10.0.0.0/16, and in the Subnet name column, clicked default, on the Edit subnet blade, I specified the settings provided and clicked on Save

Step 6

On the IP addresses tab of the Create virtual network blade, I clicked on Review + create

Step 7

I then clicked on create to complete creating the virtual network.Below is my screenshot



TASK 2

In this task,I created an application security group

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Application security groups and pressed the Enter key

Step 2

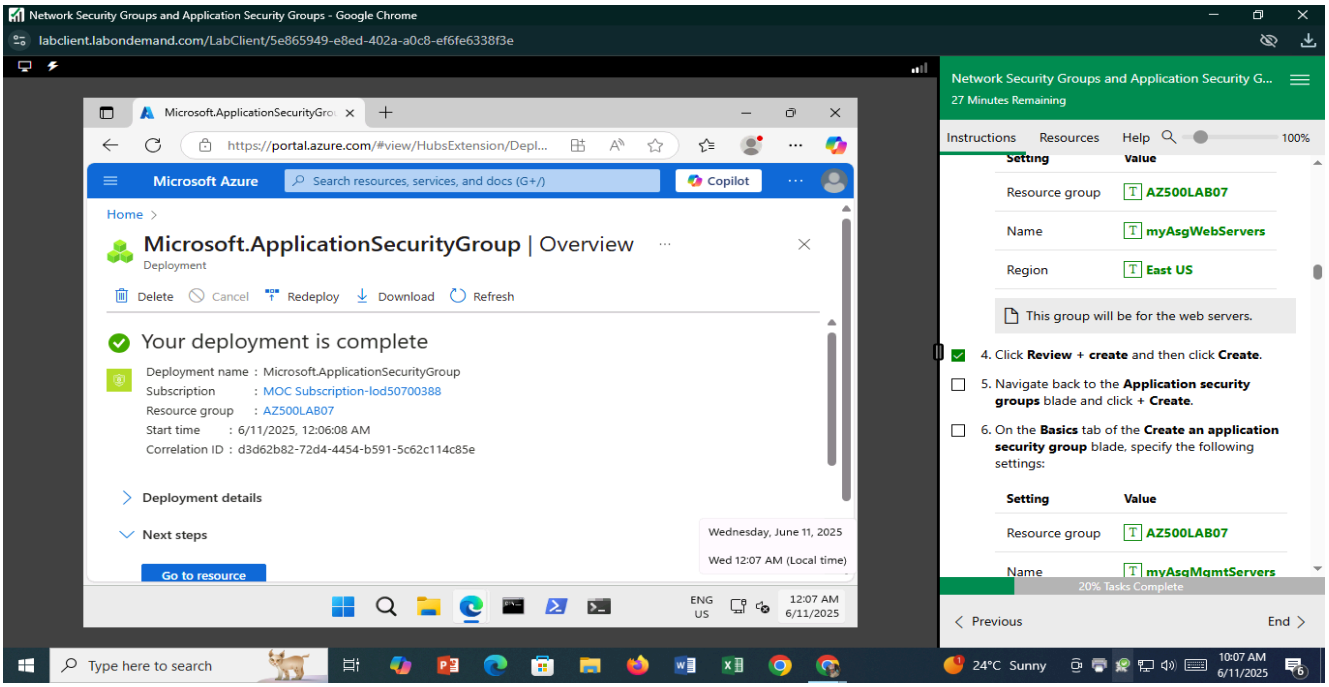
On the Application security groups blade, I clicked + Create

Step 3

On the Basics tab of the create an application security group blade, I specified the settings provided.This group was for the web servers.

Step 4

I clicked Review + create and then Create to successfully create the group. Below is my screenshot



Step 5

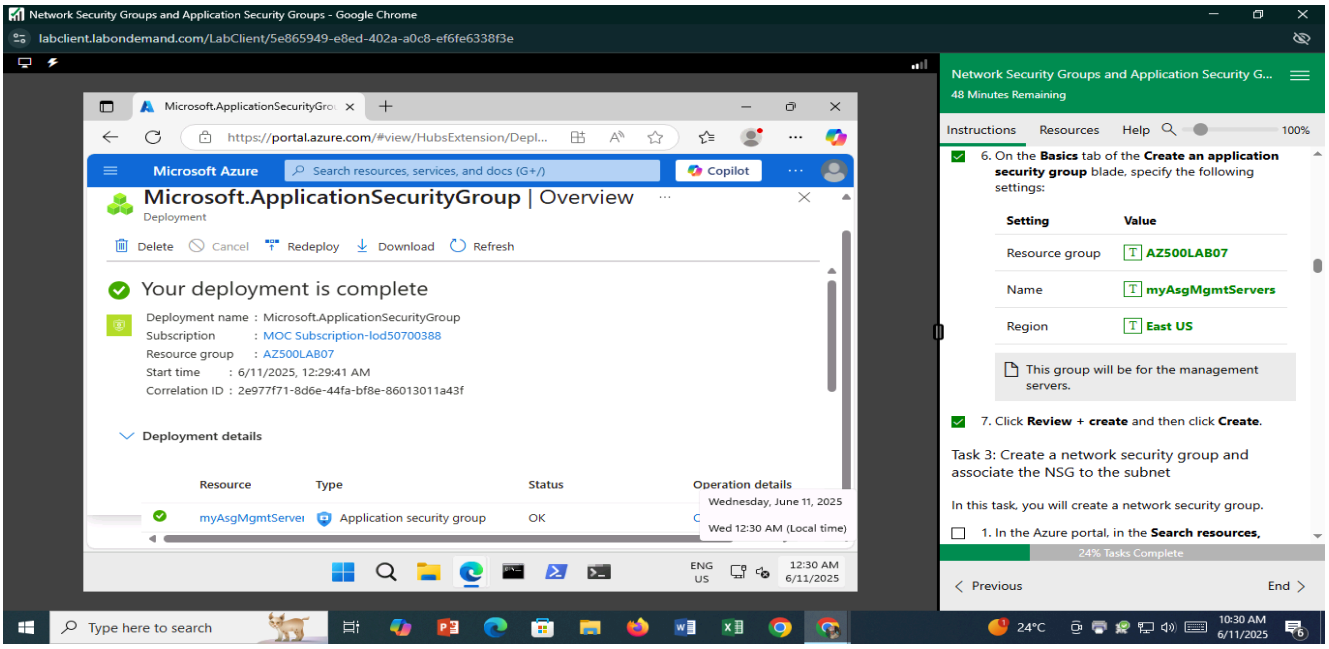
I navigated back to the Application security groups blade and clicked on + Create

Step 6

On the Basics tab of the Create an application security group blade, I specified the settings provided:This group would be for the management servers.

Step 7

I clicked on Review + create and then Create. Here is my screenshot



TASK 3

In this task,I created a network security group and associated the NSG to the subnet.

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Network security groups and pressed the Enter key.

Step 2

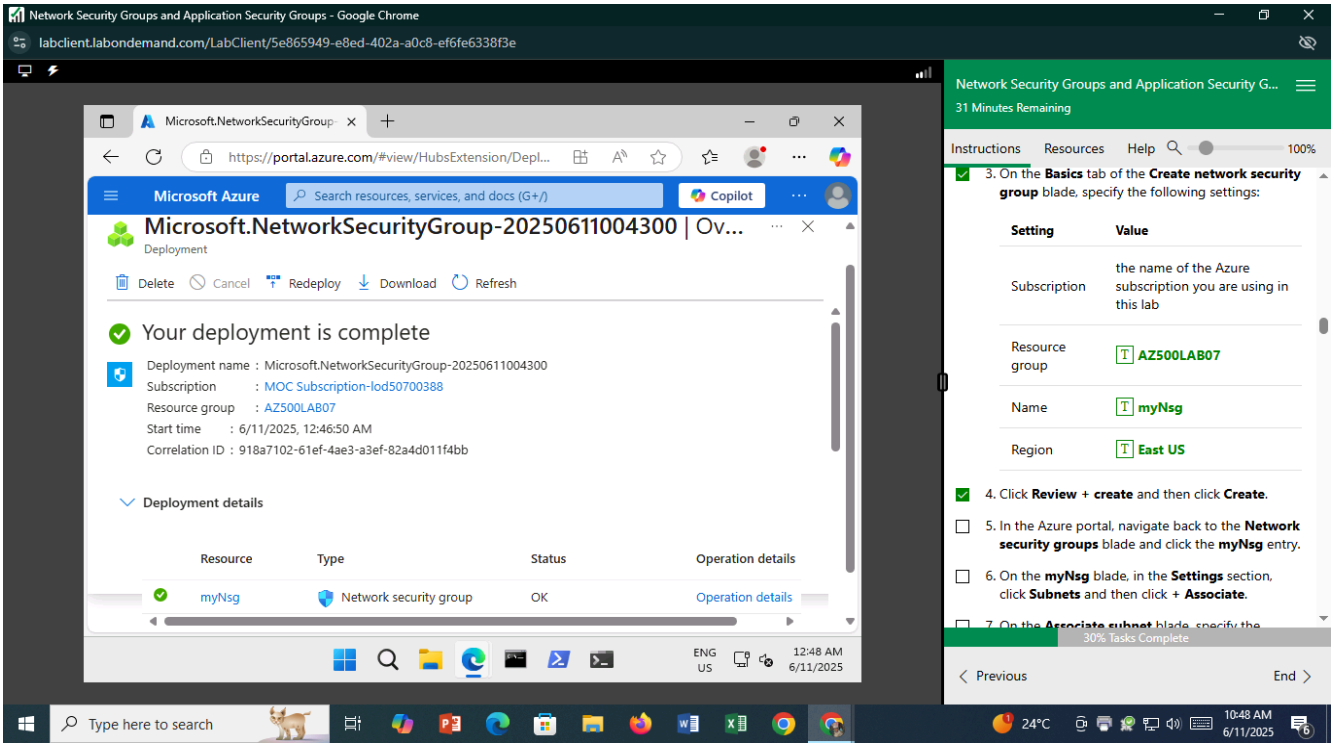
On the Network security groups blade, I clicked + Create

Step 3

On the Basics tab of the Create network security group blade, I specified the settings provided for;Subscription,Resource group,name and region.

Step 4

I then clicked on Review + create and then Create to create the network security group.Here is my screenshot



Step 5

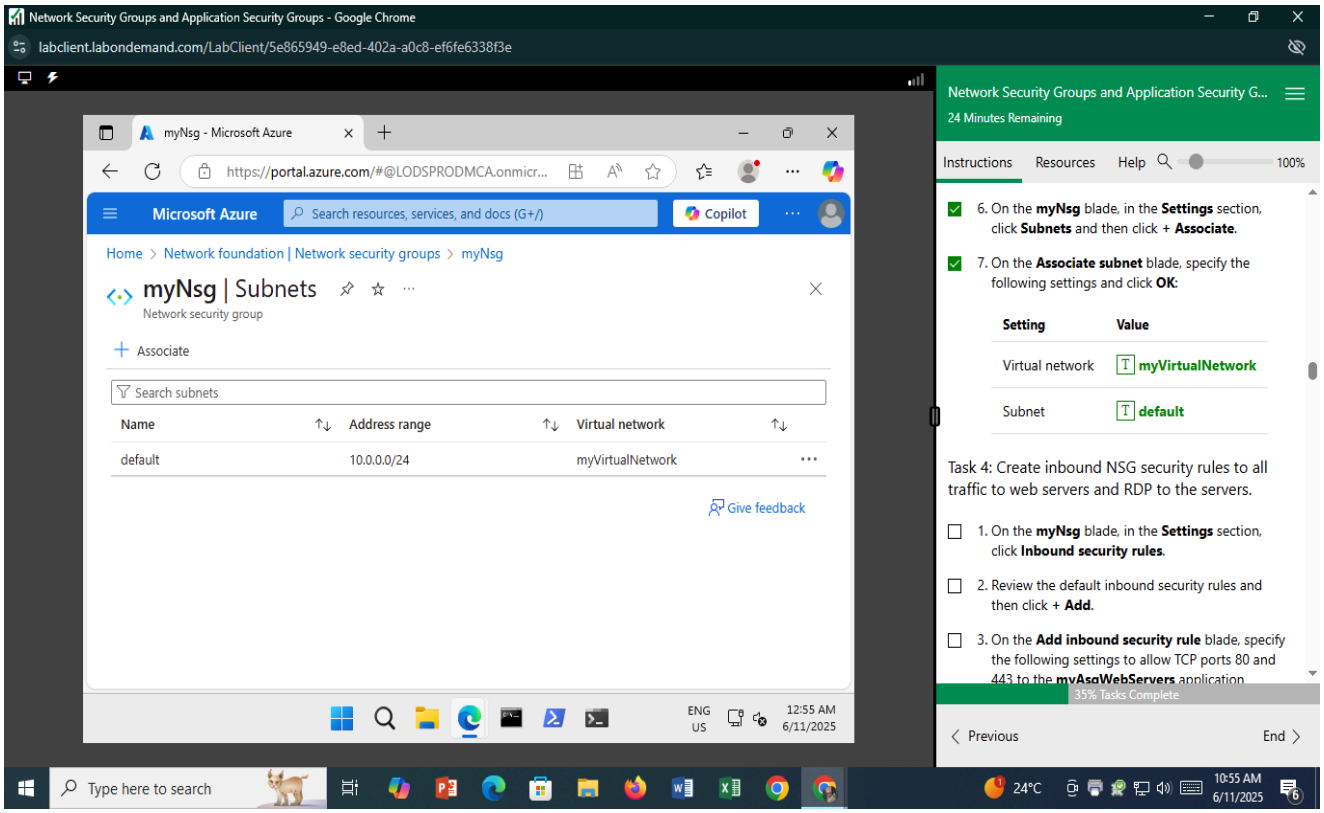
In the Azure portal, I navigated back to the Network security groups blade and clicked the myNsg entry.

Step 6

On the myNsg blade, in the Settings section, I clicked on Subnets and then clicked + Associate.

Step 7

On the Associate subnet blade, I specified the settings provided for virtual network and subnet and clicked OK.Below is my screenshot



TASK 4

In this task,I created inbound NSG security rules to all traffic to web servers and RDP to the servers.

Step 1

On the myNsg blade, in the Settings section, I clicked Inbound security rules.

Step 2

II reviewed the default inbound security rules and then clicked + Add.

Step 3

On the Add inbound security rule blade, I specified the settings provided to allow TCP ports 80 and 443 to the myAsgWebServers application security group and left all other values with their default values.Below is my screenshot

Step 4

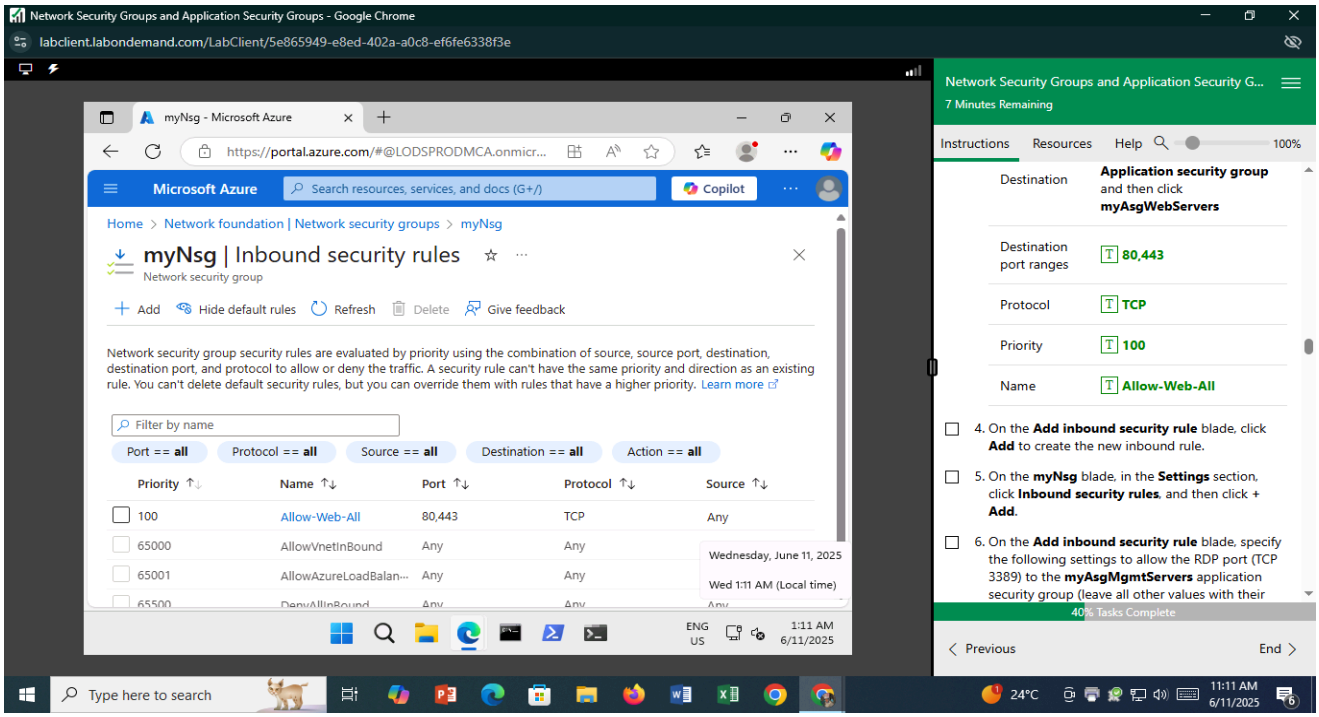
On the Add inbound security rule blade, I clicked Add to create the new inbound rule

Step 5

On the myNsg blade, in the Settings section, I clicked Inbound security rules, and then + Add

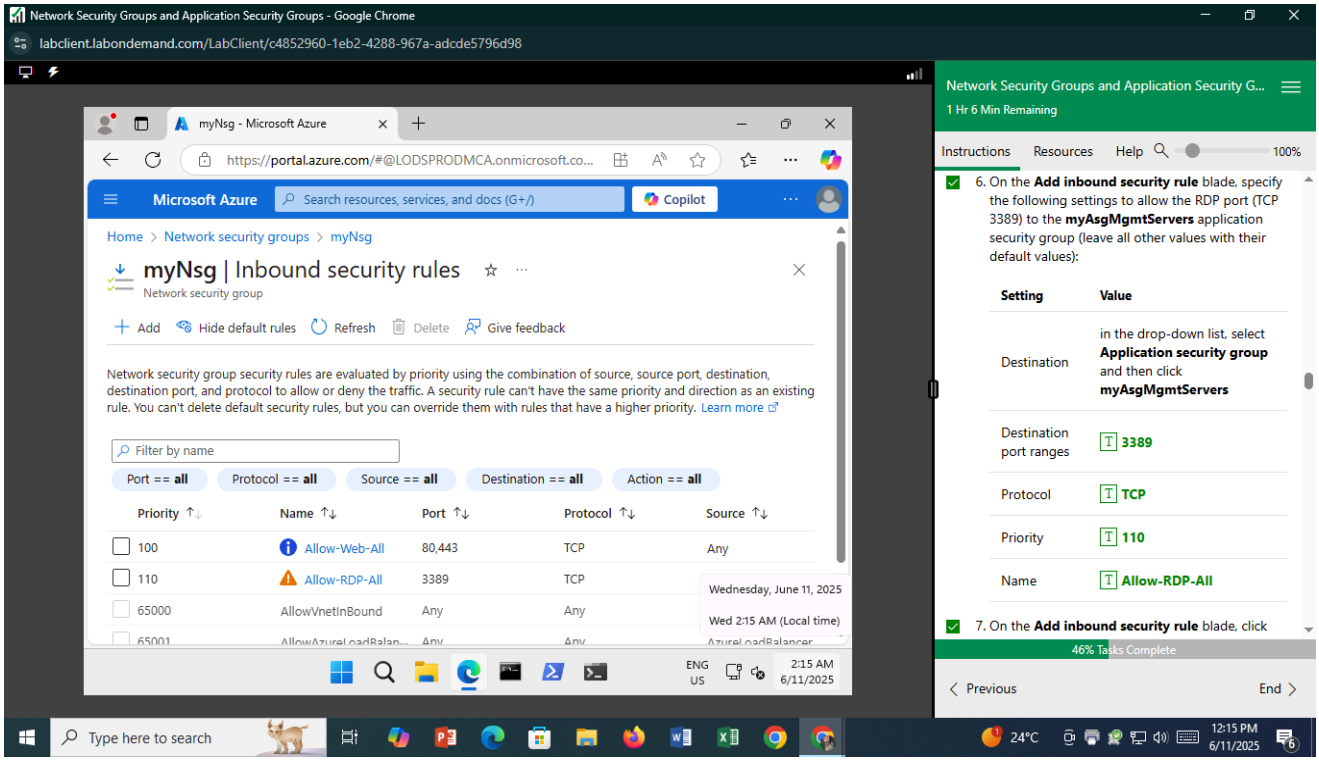
Step 6

On the Add inbound security rule blade, I specified the settings provided to allow the RDP port (TCP 3389) to the myAsgMgmtServers application security group



Step 7

On the Add inbound security rule blade, I clicked on Add to create the new inbound rule.Here is my screenshot



EXERCISE 2:DEPLOY VIRTUAL MACHINES AND TEST NETWORK FILTERS

In this exercise,I created a virtual machine to use as a web server, a virtual machine to use as a management server,associated each virtual machine's network interface to its application security group and tested the network traffic filtering.

TASK 1

In this task,I created a virtual machine to use as a web server.

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, type Virtual machines and press the Enter key.

Step 2

On the Virtual machines blade, I clicked on + Create and, in the dropdown list, clicked + Azure virtual machine.

Step 3

On the Basics tab of the Create a virtual machine blade, I specified the settings provided and left others with their default values.

Step 4

I clicked Next: Disks > and, on the Disks tab of the Create a virtual machine blade, set the OS disk type to Standard HDD and clicked Next: Networking >.

Step 5

On the Networking tab of the create a virtual machine blade, I selected the previously created network myVirtualNetwork.

Step 6

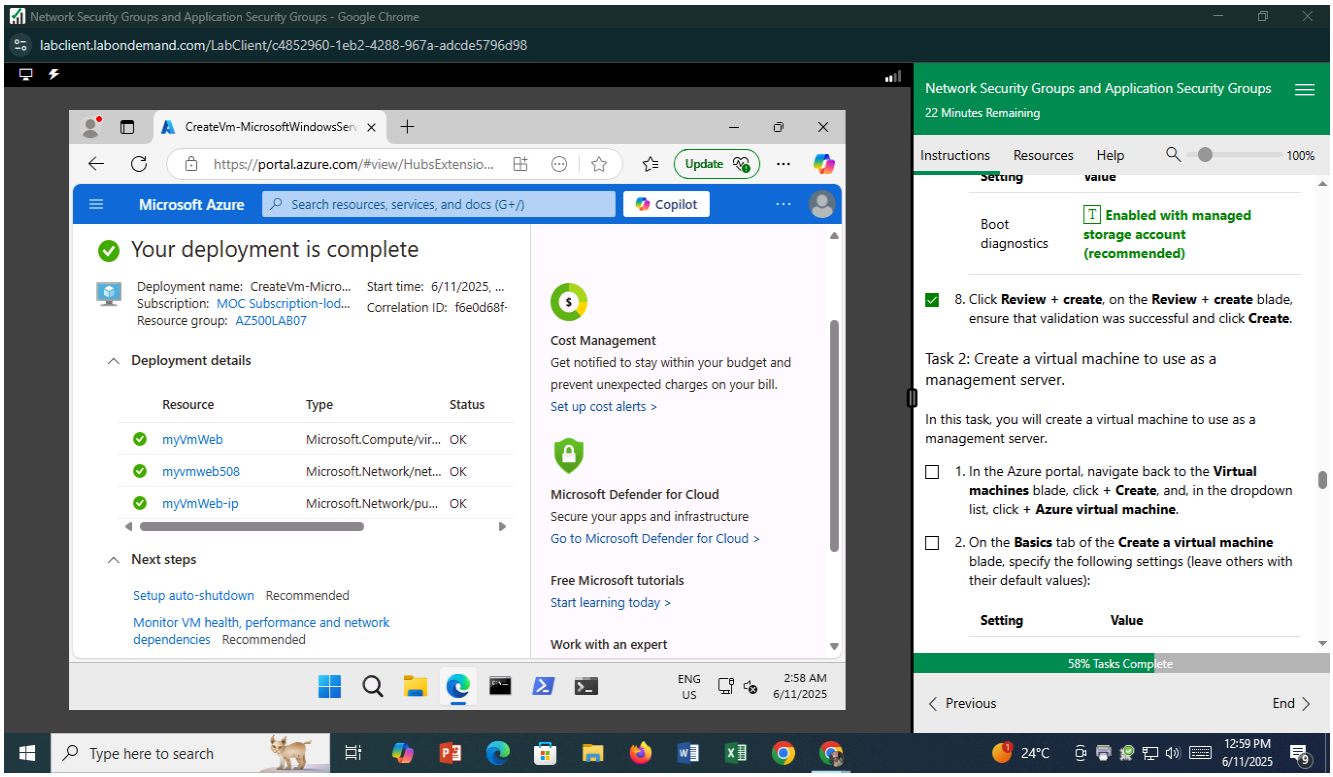
Under the NIC network security group I selected None.

Step 7

I clicked Next: Management >, then Next: Monitoring >. And On the Monitoring tab of the Create a virtual machine blade, set the .Boot diagnostics to Enabled with managed storage account(recommended)

Step 8

I clicked on Review + create, on the Review + create blade, ensured that validation was successful and clicked Create.Below is my screenshot



TASK 2

In this task,I created a virtual machine to use as a management server.

Step 1

In the Azure portal, I navigated back to the Virtual machines blade, clicked + Create, and, in the dropdown list, clicked + Azure virtual machine.

Step 2

On the Basics tab of the Create a virtual machine blade, I specified the settings provided and left others with their default values.

Step 3

I clicked Next: Disks > and, on the Disks tab, set the OS disk type to Standard HDD and clicked Next: Networking >.

Step 4

On the Networking tab of the create a virtual machine blade,I selected the previously created network myVirtualNetwork

Step 5

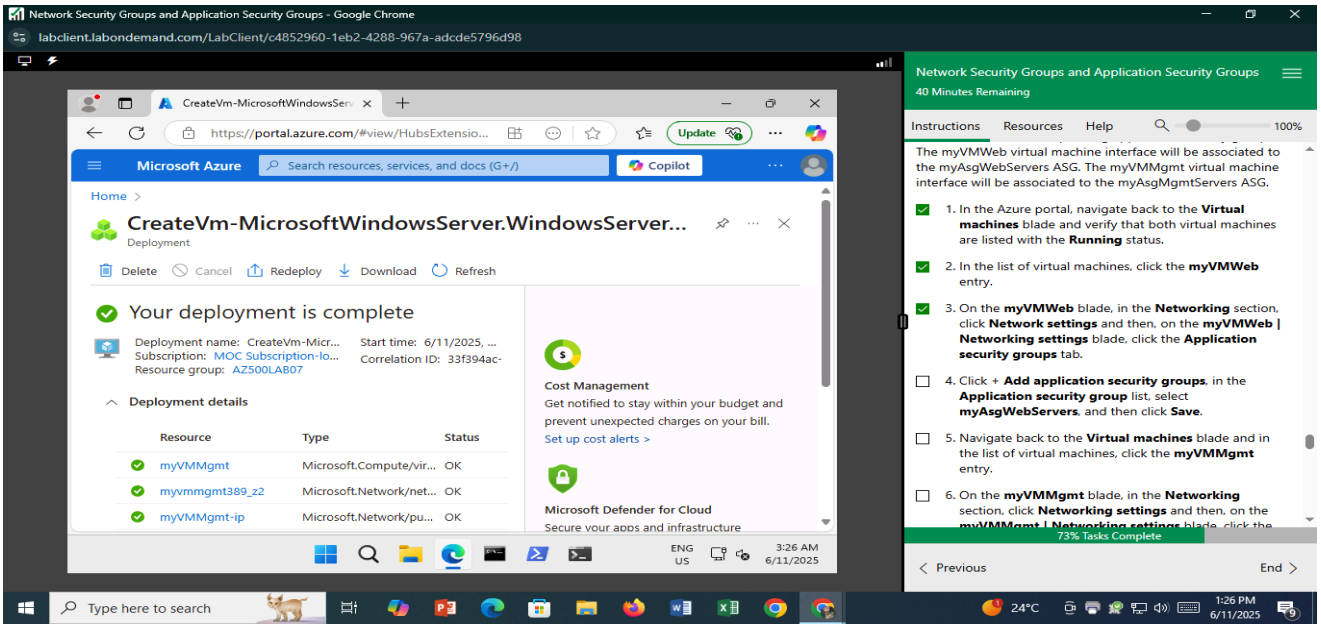
Under the NIC network security group I selected None.

Step 6

I clicked Next: Management >, then Next: Monitoring >. On the Monitoring tab of the Create a virtual machine blade, set the .Boot diagnostics to Enabled with managed storage account(recommended)

Step 7

I clicked Review + create, on the Review + create blade,I ensured that validation was successful and then clicked on Create.Here is my screenshot of successful creation of the virtual machine.

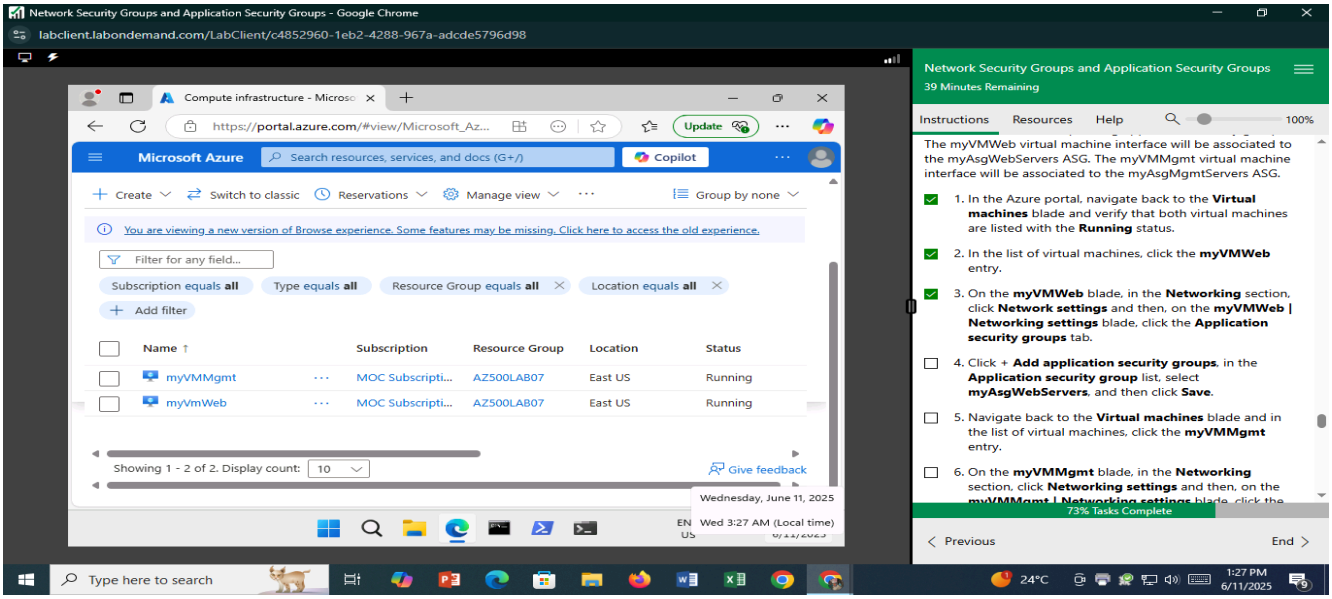


TASK 3

In this task, I associated each virtual machine's network interface with the corresponding application security group. The myVMWeb virtual machine interface to the myAsgWebServers ASG and the myVMMgmt virtual machine interface to the myAsgMgmtServers ASG.

Step 1

In the Azure portal, I navigated back to the Virtual machines blade and verified that both virtual machines are listed with the Running status.Here is my screenshot



Step 2

In the list of virtual machines, I clicked the myVMWeb entry

Step 3

On the myVMWeb blade, in the Networking section, I clicked on Network settings and then, on the myVMWeb | Networking settings blade, clicked the Application security groups tab.

Step 4

I clicked + Add application security groups, in the Application security group list, selected myAsgWebServers, and then clicked Save.

Step 5

I navigated back to the Virtual machines blade and in the list of virtual machines, clicked the myVMMgmt entry.

Step 6

On the myVMMgmt blade, in the Networking section,I clicked Networking settings and then, on the myVMMgmt | Networking settings blade, clicked the Application security groups tab.

Step 7

I clicked the + Add application security groups, in the Application security group list, selected myAsgMgmtServers, and then clicked on Save.

TASK 4

In this task, I tested the network traffic filters.

Step 1

I navigate back to the myVMMgmt virtual machine blade.

Step 2

On the myVMMgmt blade, I clicked on Connect and RDP in the drop down menu.

Step 3

I clicked Download RDP File and used it to connect to the myVMMgmt Azure VM via Remote Desktop. Below is a screenshot of my successful connection.

On the Run Command Script pane, run the following to install the Web server role on myVmWeb

Step 4

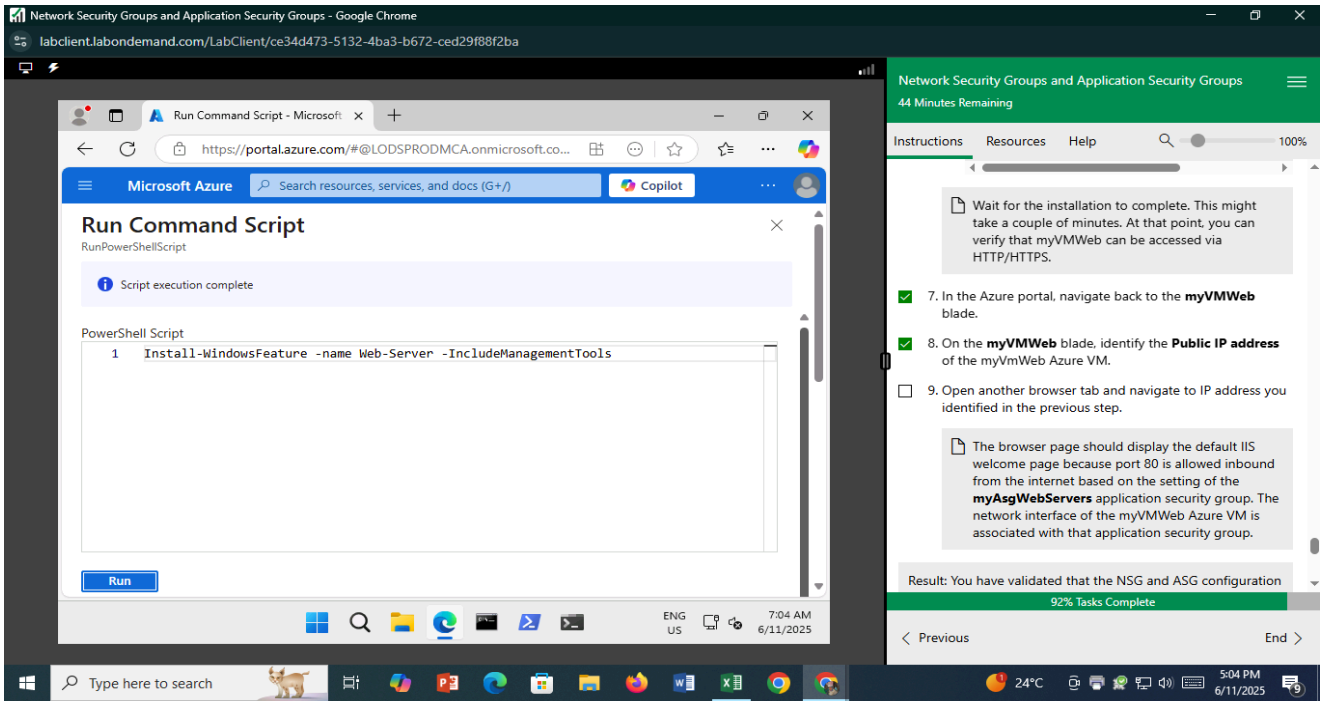
In the Azure portal, I navigated to the myVMWeb virtual machine blade

Step 5

On the myVMWeb blade, in the Operations section, I clicked Run command and RunPowerShellScript

Step 6

On the Run Command Script pane, I ran the command provided to install the Web server role on myVmWeb.Below is my screenshot

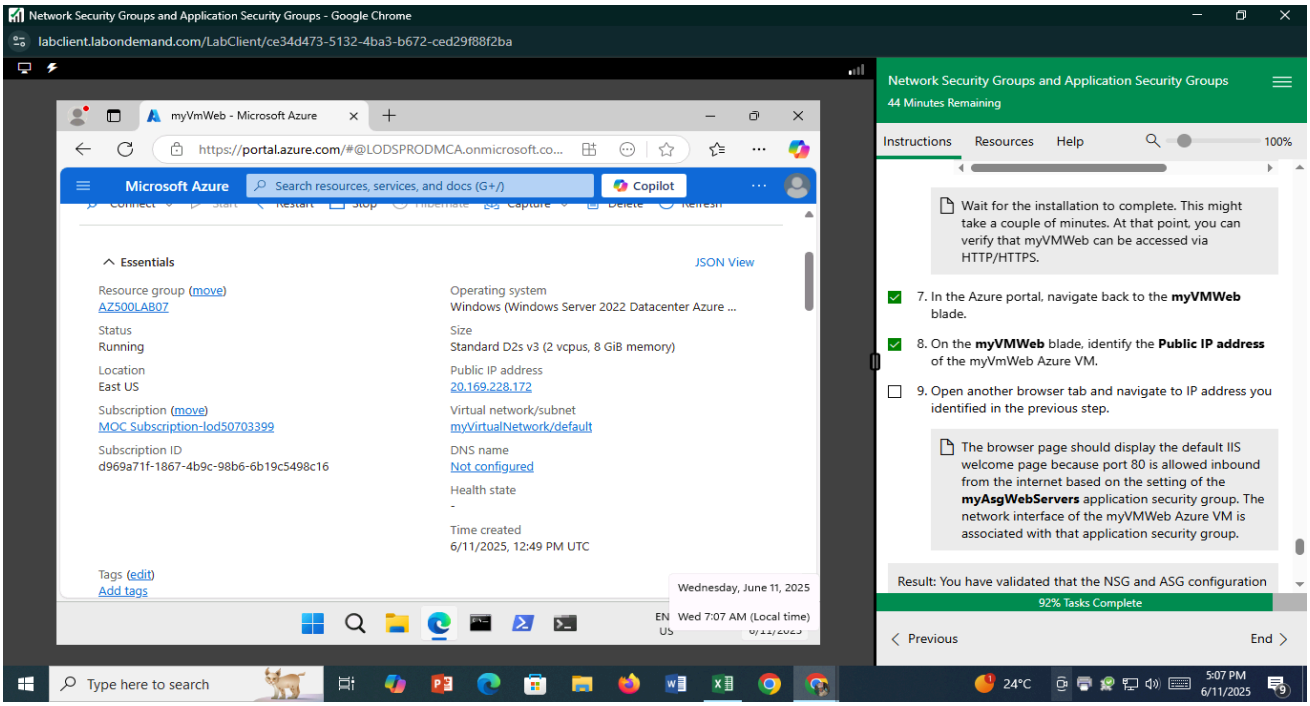


Step 7

In the Azure portal, I navigated back to the myVMWeb blade.

Step 8

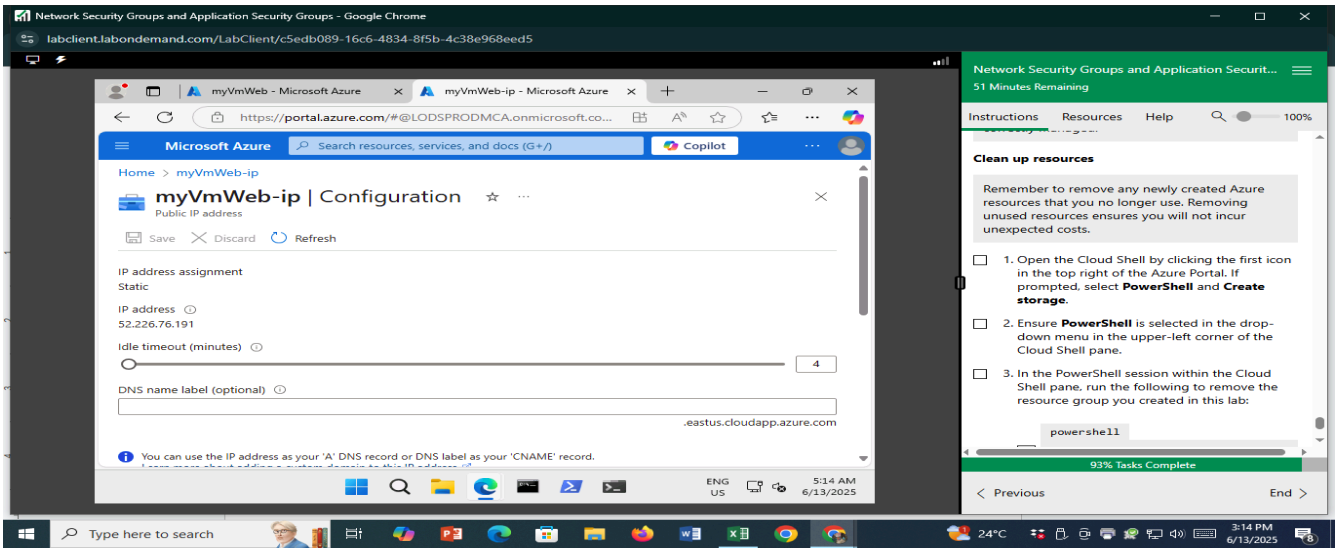
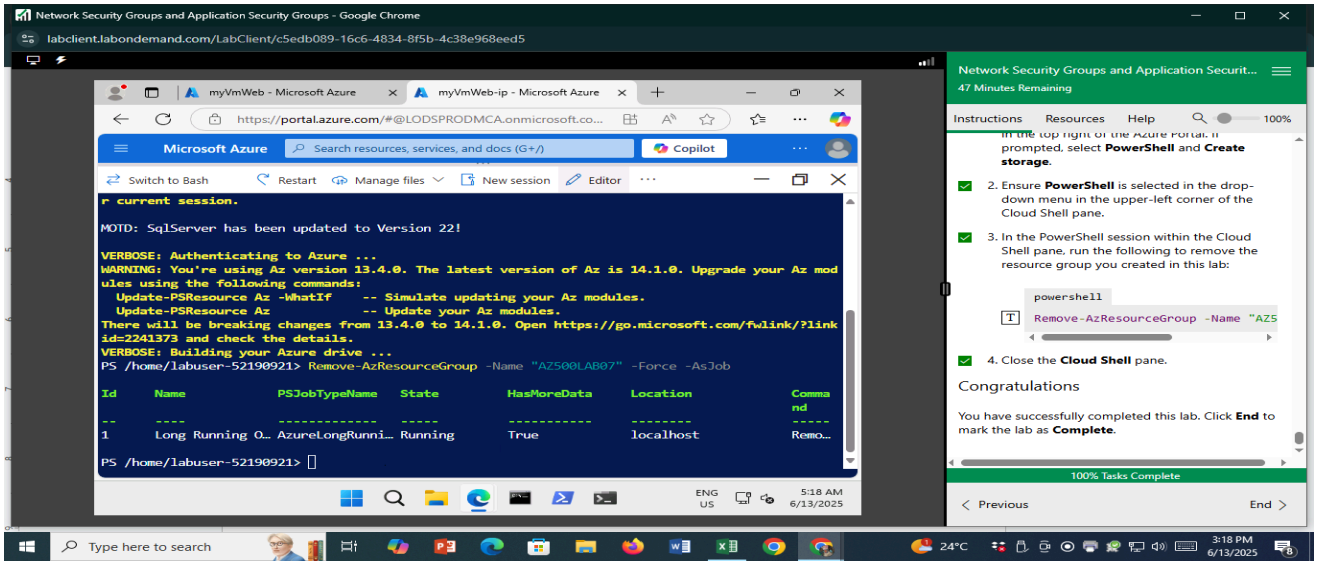
On the myVMWeb blade, I identified the Public IP address of the myVmWeb Azure VM.Here is my screenshot



Step 9

I opened another browser tab and navigated to the IP address identified in the previous step.The browser page displayed the default IIS welcome page because port 80 is allowed inbound from the internet based on the setting of the myAsgWebServers application security group.

Here is my screenshot



I cleaned up my resources and completed the lab.Here is my screenshot